

IMPLICITLY RESTARTED ARNOLDI METHODS AND SUBSPACE ITERATION*

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Abstract. This goal of this paper is to present an elegant relationship between an implicitly restarted Arnoldi method (IRAM) and nonstationary (subspace) simultaneous iteration. This relationship allows the geometric convergence theory developed for nonstationary simultaneous iteration due to Watkins and Elsner [*Linear Algebra Appl.*, 143 (1991), pp. 19–47] to be used for analyzing the rate of convergence of an IRAM. We also comment on the relationship with other restarting schemes. A set of experiments demonstrates that implicit restarted methods can converge at a much faster rate than simultaneous iteration when iterating on a subspace of equal dimension.

Key words. simultaneous iteration, Arnoldi reduction, Schur decomposition, restarting, eigenvalues

AMS subject classifications. 65F15, 65G05

PII. S0895479899358595

1. Introduction. A classical method of solution for the large-scale eigenvalue problem is simultaneous iteration [6, 9, 25, 26, 28, 36, 38]. Simultaneous iteration was originally introduced by Bauer [7], who called the method *Treppeniteration* (staircase iteration). The iteration is a straightforward method for computing the eigenvalues of largest modulus of a matrix $\mathbf{A}$ and is a generalization of the power method in that a subspace of size larger than one is employed. Unfortunately, the rate of convergence of simultaneous iteration is often prohibitive for many large-scale eigenvalue problems.

The goal and contribution of this paper is the derivation of an elegant relationship between an implicitly restarted Arnoldi method (IRAM) and nonstationary subspace or simultaneous iteration. We show that an IRAM, including its block extension, is nonstationary simultaneous iteration in disguise. The relationship with nonstationary simultaneous iteration is demonstrated by exploiting a well-known connection [24, 37, 39] with the QR algorithm. By appealing to the results in [40], possible rates of convergence for an IRAM are established. Moreover, we show how an IRAM computes a nested sequence of approximations for the partial Schur decomposition associated with the dominant invariant subspace of a matrix. Numerical experiments show that an IRAM can converge significantly faster than simultaneous iteration when acting on subspaces of equal dimension.

We remark that the relationship between an IRAM and simultaneous iteration presented also applies to all other restarted Arnoldi methods that employ the same initial Krylov space and restart polynomial. The elegant relationship with simultaneous iteration provided by implicit restarting, however, also provides a robust and stable algorithm. This robustness and stability is a direct result of an exclusive reliance on

*Received by the editors July 12, 1999; accepted for publication (in revised form) by Z. Strakoš May 18, 2001; published electronically November 13, 2001. This work was supported by the Department of Energy (contracts DE-FG0f-91ER25103 and W-31-109-Eng-38). The U.S. Government retains a nonexclusive, royalty-free license to publish or reproduce the published form of this contribution, or allow others to do so, for U.S. Government purposes. Copyright is owned by SIAM to the extent not limited by these rights.

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unitary transformations. Golub and Wilkinson [11] examine the many practical difficulties involved when computing invariant subspaces. They conclude that working with an orthonormal basis of approximate Schur vectors is a better-behaved numerical process. Within the context of simultaneous iteration, Stewart [38] also arrives at the same conclusion.

Section 2 discusses the eigenvalue problem and Schur decompositions. Section 3 introduces the Arnoldi reduction and its computation. The relationship between an IRAM and nonstationary simultaneous iteration is derived in section 4. Section 5 describes a geometric convergence theory of an IRAM. Section 6 discusses some practical issues related to the convergence of an IRAM. The results of some numerical experiments comparing simultaneous iteration with an IRAM are given in section 7.

We conclude this section with the basic notation to be used in this article. We employ Householder notational conventions. Capital and lowercase letters denote matrices and vectors, respectively, while lowercase Greek letters denote scalars.

The transpose of a vector $\mathbf{x}$ is denoted by $\mathbf{x}^T$, and the complex conjugate of $\mathbf{x}^T$ is denoted by $\mathbf{x}^H$. The norms used are the Euclidean and Frobenius, denoted by $\|\cdot\|$ and $\|\cdot\|_F$, respectively. The range of a matrix $\mathbf{A}$ is denoted by $\mathcal{R}(\mathbf{A})$.

A matrix of lower bandwidth b will be called a banded upper Hessenberg matrix. We drop “upper” when the context is clear. Omission of the word *band* implies that the block size is one. We say that a banded Hessenberg matrix is unreduced if all the elements on the b th subdiagonal are nonzero.

2. The eigenvalue problem. Let $\mathbf{A}$ be a complex matrix of order n . We are interested in computing the $k \ll n$ dominant (those of largest magnitude) eigenvalues and associated invariant subspace of

$$(2.1) \quad \mathbf{A}\mathbf{x} = \lambda\mathbf{x}.$$

The eigenvalues and eigenvectors of $\mathbf{A}$ are denoted by λ_j and $\mathbf{x}_j$, respectively, for $j = 1, \dots, n$. For the remainder of our article, we assume that $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$ and $|\lambda_k| > |\lambda_{k+1}|$.

The following decomposition proves central to the eigenvalue algorithms considered in this article. The decomposition’s value is in providing us with a canonical form for which stable algorithms may be developed. For us, a stable algorithm computes the exact Schur decomposition of a nearby matrix.

THEOREM 2.1 (Schur decomposition). *If $\mathbf{A} \in \mathbf{C}^{n \times n}$, then there exists a unitary $\mathbf{Z} \in \mathbf{C}^{n \times n}$ such that*

$$(2.2) \quad \mathbf{Z}^H \mathbf{A} \mathbf{Z} = \mathbf{T},$$

where $\mathbf{T}$ is an upper triangular matrix. The eigenvalues can appear in any order along the diagonal.

Proof. See [10, p. 313]. $\square$

Let $\mathbf{D}$ be a diagonal unitary matrix. Then $(\mathbf{ZD})^H \mathbf{A} \mathbf{ZD} = \mathbf{D}^H \mathbf{T} \mathbf{D}$ has diagonal blocks equal to those of $\mathbf{T}$. Thus, apart from the eigenvalues of multiplicity larger than one, the decomposition is essentially unique, given some ordering of the eigenvalues. Denote the leading principal matrix of order k of $\mathbf{T}$ by $\hat{\mathbf{T}}_k$. Let $\mathbf{Z}_k$ be the corresponding columns of $\mathbf{Z}$. Then $\mathbf{A} \mathbf{Z}_k = \mathbf{Z}_k \hat{\mathbf{T}}_k$ is a *partial* Schur decomposition of $\mathbf{A}$ of order k . When $\mathbf{A}$ is Hermitian, $\mathbf{T}$ is a diagonal matrix, and hence the eigenvalues are real numbers.

The full decomposition is computed by the practical QR algorithm in the EISPACK [33] and LAPACK [1] software packages. Instead simultaneous iteration attempts to compute a partial Schur decomposition for $\mathbf{A}$ with the dominant eigenvalues located on the diagonal of $\hat{\mathbf{T}}_k$.

3. Partial reduction to band Hessenberg form. The first step of the practical QR algorithm is to reduce $\mathbf{A}$ to upper Hessenberg form via Householder transformations. This is done because each step of the QR iteration performed on a Hessenberg matrix involves only order n^2 work. This is in contrast to the order n^3 work that would be required during each step of a QR iteration on $\mathbf{A}$ (assuming that $\mathbf{A}$ is a dense matrix).

Unfortunately, for large eigenvalue problems, Householder transformations cannot be used as they destroy any sparsity or structure in $\mathbf{A}$. We say an eigenvalue problem is large if the dense QR algorithm is prohibitive in storage and/or efficiency. Instead, the Arnoldi reduction [2] only requires knowledge of $\mathbf{A}$ through a matrix-vector product. Moreover, it allows us to sequentially reduce $\mathbf{A}$ to upper Hessenberg form, producing the leading portion of an upper Hessenberg matrix at every step. When the matrix $\mathbf{A}$ is Hermitian, the Lanczos reduction [16] is recovered.

Since our concern is in the solution of eigenvalue problems where $\mathbf{A}$ is not only large but also expensive to apply, block Arnoldi reductions [30, 31] are considered. In many instances, the cost of computing a few matrix-vector products is commensurate with that of one matrix-vector product. Moreover, in exact arithmetic, an unblocked Arnoldi reduction cannot detect the multiplicity of an eigenvalue. A blocksize enables the reduction to compute multiplicities less than or equal to the blocksize.

Let $b > 0$, an integer, be the blocksize and let $\mathbf{E}_j \equiv [\mathbf{e}_{(j-1)b+1} \cdots \mathbf{e}_{jb}]$ where the j th canonical basis vector is denoted by $\mathbf{e}_j$. We say that

$$(3.1) \quad \mathbf{A}\mathbf{V}_j = \mathbf{V}_j\mathbf{H}_j + \mathbf{F}_j\mathbf{E}_j^T$$

is a block Arnoldi reduction of length j when $\mathbf{V}_j^H\mathbf{A}\mathbf{V}_j = \mathbf{H}_j$ is a banded upper Hessenberg matrix, $\mathbf{V}_j^H\mathbf{V}_j = \mathbf{I}_{j \cdot b}$, and $\mathbf{V}_j^H\mathbf{F}_j = \mathbf{0}$. The $j \cdot b$ columns of $\mathbf{V}_j$ are an orthonormal basis for the block Krylov subspace

$$\mathcal{K}_j^b(\mathbf{A}, \mathbf{U}_1) \equiv \text{Span}\{\mathbf{U}_1, \mathbf{A}\mathbf{U}_1, \dots, \mathbf{A}^{j-1}\mathbf{U}_1\},$$

where $\mathbf{U}_1$ is a full rank matrix with b columns. Note that if $\mathbf{A} = \mathbf{A}^H$, then $\mathbf{H}_j$ is a block tridiagonal matrix.

In order to introduce notation that will be needed, we rewrite (3.1) as

$$\begin{aligned} \mathbf{A}\mathbf{V}_j = [\mathbf{U}_1 \cdots \mathbf{U}_j] & \begin{bmatrix} \mathbf{G}_{1,1} & \cdots & \cdots & \mathbf{G}_{1,j} \\ \mathbf{G}_{2,1} & \ddots & \vdots & \vdots \\ \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \cdots & \mathbf{G}_{j,j-1} & \mathbf{G}_{j,j} \end{bmatrix} \\ & + \mathbf{U}_{j+1}\mathbf{G}_{j+1,j}\mathbf{E}_j^T, \end{aligned}$$

where $\mathbf{U}_{j+1}\mathbf{G}_{j+1,j}$ is the QR factorization of $\mathbf{F}_j$. The reduction is advanced by one step (or its length incremented) by the following three operations:

1. $\mathbf{W} = \mathbf{A}\mathbf{U}_{j+1}$.
2. $\mathbf{G}_{i,j+1} = \mathbf{U}_i^H\mathbf{W}$ $i = 1, \dots, j+1$.
3. $\mathbf{F}_{j+1} = \mathbf{W} - \sum_{i=1}^{j+1} \mathbf{U}_i\mathbf{G}_{i,j+1}$.

A direct implementation of the second and third steps will not, in general, produce an orthogonal set of Arnoldi vectors. We refer the reader to [17] for details on an efficient and robust implementation. See [12] for references and information on a block Lanczos reduction implemented via a three-term block recurrence.

3.1. Computing an approximate partial Schur decomposition. Suppose that $\mathbf{H}_j \mathbf{Z} = \mathbf{Z} \mathbf{T}$ is a Schur decomposition ordered so that the eigenvalues of $\mathbf{T}$ are in descending order of magnitude along the diagonal. Postmultiplying (3.1) with $\mathbf{Z}$ gives

$$(3.2) \quad \|\mathbf{A} \mathbf{V}_j \mathbf{Z} - \mathbf{V}_j \mathbf{Z} \mathbf{T}\| = \|\mathbf{F}_j \mathbf{E}_j^T \mathbf{Z}\| = \|\mathbf{G}_{j+1,j} \mathbf{E}_j^T \mathbf{Z}\|.$$

Because $\mathbf{E}_j^T \mathbf{Z}$ is a matrix with $j \cdot b$ columns consisting of the last b rows of $\mathbf{Z}$, the quality of an approximate partial Schur decomposition is determined because

$$\|\mathbf{F}_j \mathbf{E}_j^T \mathbf{Z} [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \cdots \quad \mathbf{e}_k]\| \equiv \|\mathbf{F}_j \mathbf{E}_j^T \mathbf{Z}_k\|.$$

If this quantity is small, then an approximate partial Schur decomposition of order k for the dominant invariant subspace is computed. We now establish this assertion. Suppose that $\mathbf{H}_j \mathbf{Z}_k = \mathbf{Z}_k \hat{\mathbf{T}}_k$ is an order k partial Schur decomposition for $\mathbf{H}_j$; then

$$(\mathbf{A} + \mathbf{M}) \mathbf{V}_j \mathbf{Z}_k = \mathbf{V}_j \mathbf{Z}_k \hat{\mathbf{T}}_k,$$

where

$$\|\mathbf{M}\| = \|\mathbf{F}_j \mathbf{E}_j^T \mathbf{Z}_k (\mathbf{V}_j \mathbf{Z}_k)^H\| \leq \|\mathbf{G}_{j+1,j} \mathbf{E}_j^T \mathbf{Z}_k\|,$$

implying that we have computed an exact partial Schur decomposition for a matrix near $\mathbf{A}$.

If approximate eigenvectors are of interest, they can be computed from the partial Schur decomposition $\mathbf{H}_j \mathbf{Z}_k = \mathbf{Z}_k \hat{\mathbf{T}}_k$ because

$$(3.3) \quad \mathbf{A} \mathbf{V}_j \mathbf{Z}_k \mathbf{y} - \mathbf{V}_j \mathbf{Z}_k \mathbf{y} \theta = \mathbf{F}_j \mathbf{E}_j^T \mathbf{Z}_k \mathbf{y},$$

where $\hat{\mathbf{T}}_k \mathbf{y} = \mathbf{y} \theta$. We call $\mathbf{V}_j \mathbf{Z}_k \mathbf{y}$ a Ritz vector and θ a Ritz value. Note that the first Schur vector is always an eigenvector.

4. Connection with nonstationary simultaneous iteration. There is a well-known connection between nonstationary simultaneous iteration and the QR algorithm. This section will show a similar relationship between an IRAM and nonstationary simultaneous iteration. The following elementary but technical result is needed for this relationship. The result is a generalization of the special case $\psi(\lambda) = \lambda$ shown in [19]. A similar result was proved in Lemma 1 of [23] for the Lanczos reduction.

LEMMA 4.1. *Suppose that an integer p satisfies $1 \leq p < m$, and let $r = m - p$. Let $\mathbf{A} \mathbf{V}_m = \mathbf{V}_m \mathbf{H}_m + \mathbf{F}_m \mathbf{E}_m^T$ be a length $r + p$ Arnoldi reduction, where $\mathbf{H}_m$ is an unreduced band upper Hessenberg matrix. If*

$$\psi_p(\lambda) = \prod_{i=1}^p (\lambda - \tau_i),$$

then

$$(4.1) \quad \psi_p(\mathbf{A}) \mathbf{V}_m = \mathbf{V}_m \psi_p(\mathbf{H}_m) + \sum_{j=1}^p \psi_{j+1,p}(\mathbf{A}) \mathbf{F}_m \mathbf{E}_m^T \psi_{j-1}(\mathbf{H}_m),$$

where $\psi_{j,p}(\lambda) = \prod_{i=j}^p (\lambda - \tau_i)$.
Moreover,

$$(4.2) \quad \psi_p(\mathbf{A})\mathbf{V}_r = \mathbf{V}_m\psi_p(\mathbf{H}_m) \begin{bmatrix} \mathbf{E}_1 & \cdots & \mathbf{E}_r \end{bmatrix}.$$

Proof. The proof is by mathematical induction. Define $m \equiv r + p$. The subscripts are suppressed on $\mathbf{V}_m$ and $\mathbf{H}_m$ for the proof. Since $\psi_1(\mathbf{A})\mathbf{V} = \mathbf{V}\psi_1(\mathbf{H}) + \mathbf{F}_m\mathbf{E}_m^T$, where $\psi_1(\lambda) = \lambda - \tau_1$, the base case for $p = 1$ is established. Assume the lemma's truth for polynomials $\psi_j(\lambda)$ of degree $j \leq p$. Let $\psi_{p+1}(\lambda) = (\lambda - \tau_{p+1})\psi_p(\lambda)$. Using the induction hypothesis leads to

$$\begin{aligned} \psi_{p+1}(\mathbf{A})\mathbf{V} &= (\mathbf{A} - \tau_{p+1}\mathbf{I})\psi_p(\mathbf{A})\mathbf{V} \\ &= (\mathbf{A} - \tau_{p+1}\mathbf{I}) \left\{ \mathbf{V}\psi_p(\mathbf{H}) + \sum_{j=1}^p \psi_{j+1,p}(\mathbf{A})\mathbf{F}_m\mathbf{E}_m^T\psi_{j-1}(\mathbf{H}) \right\} \\ &= \mathbf{V}(\mathbf{H} - \tau_{p+1}\mathbf{I})\psi_p(\mathbf{H}) + \mathbf{F}_m\mathbf{E}_m^T\psi_p(\mathbf{H}) \\ &\quad + (\mathbf{A} - \tau_{p+1}\mathbf{I}) \sum_{j=1}^p \psi_{j+1,p}(\mathbf{A})\mathbf{F}_m\mathbf{E}_m^T\psi_{j-1}(\mathbf{H}) \\ &= \mathbf{V}\psi_{p+1}(\mathbf{H}) + \sum_{j=1}^{p+1} \psi_{j+1,p+1}(\mathbf{A})\mathbf{F}_m\mathbf{E}_m^T\psi_{j-1}(\mathbf{H}), \end{aligned}$$

which establishes (4.1).

Since $\mathbf{H}$ is unreduced, $\psi_{j-1}(\mathbf{H})$ is a band Hessenberg matrix of lower bandwidth $(j-1) \cdot b$. Hence $\mathbf{E}_i^T\psi_{j-1}(\mathbf{H})\mathbf{E}_l = \mathbf{0}$ for $l = 1, \dots, m-j$, and the last matrix on the right-hand side of (4.1) is zero through its first $r \cdot b$ columns and (4.2) is established. $\square$

In other words, (4.2) shows that $\psi_p(\mathbf{A})$ applied to the first $r \cdot b$ columns of $\mathbf{V}_m$ is equivalent to the basis representation given by the first $r \cdot b$ columns of $\mathbf{V}_m\psi_p(\mathbf{H}_m)$.

Suppose that p steps of the QR algorithm are performed on $\mathbf{H}_m$ with $p < m$ shifts $\tau_1, \dots, \tau_p$ resulting in

$$(4.3) \quad \mathbf{H}_m\mathbf{W} = \mathbf{W}\mathbf{H}_m^+.$$

Note that $\mathbf{W}$ is a Hessenberg matrix of lower bandwidth $p \cdot b$ because it is a product of p unitary matrices each of lower bandwidth b (each computed during a step of the QR algorithm). If $\mathbf{W}_1$ denotes the initial $r \cdot b$ columns of $\mathbf{W}$, $\mathbf{W}_2$ denotes the next b columns, and $\mathbf{W}_3$ denotes the remaining columns, equating the first $r \cdot b$ columns of (4.3) results in

$$(4.4) \quad \mathbf{H}_m\mathbf{W}_1 \equiv \begin{bmatrix} \mathbf{W}_1 & \mathbf{W}_2 & \mathbf{W}_3 \end{bmatrix} \begin{bmatrix} \mathbf{H}_r^+ \\ \mathbf{G}_{r+1,r}^+\mathbf{E}_r^T \\ \mathbf{0} \end{bmatrix}.$$

Postmultiplying (3.1) with $\mathbf{W}_1$ and using (4.4), we obtain

$$\begin{aligned} (4.5) \quad \mathbf{A}\mathbf{V}_m\mathbf{W}_1 &= \mathbf{V}_m\mathbf{H}_m\mathbf{W}_1 + \mathbf{F}_m\mathbf{E}_m^T\mathbf{W}_1 \\ &= \mathbf{V}_m\mathbf{W}_1\mathbf{H}_r^+ + \mathbf{V}_m\mathbf{W}_2\mathbf{G}_{r+1,r}^+\mathbf{E}_r^T + \mathbf{F}_m\mathbf{E}_m^T\mathbf{W}_1 \\ &\equiv \mathbf{V}_r^+\mathbf{H}_r^+ + \mathbf{F}_r^+\mathbf{E}_r^T, \end{aligned}$$

where

$$(4.6) \quad \mathbf{V}_r^+ \equiv \mathbf{V}_m \mathbf{W}_1 \quad \text{and} \quad \mathbf{F}_r^+ \equiv \mathbf{V}_m \mathbf{W}_2 \mathbf{G}_{r+1,r}^+ + \mathbf{F}_m \mathbf{E}_m^T \mathbf{W}_1 \mathbf{E}_r.$$

Note the use of the identity $\mathbf{E}_m^T \mathbf{W}_1 = [\mathbf{0} \quad \cdots \quad \mathbf{0} \quad \mathbf{E}_m^T \mathbf{W}_1 \mathbf{E}_r]$ in (4.6).

The following theorem establishes a direct relationship between nonstationary simultaneous iteration and the QR algorithm. The theorem is a partial or truncated version of Theorem 3.1 in [37, p. 353].

THEOREM 4.2. *Assume the hypothesis of Lemma 4.1 and the notation in (4.4)–(4.6). If the QR algorithm computed on $\mathbf{H}_m$ with the p shifts $\tau_1, \dots, \tau_p$ results in $\mathbf{H}_m \mathbf{W} = \mathbf{W} \mathbf{H}_m^+$, then*

$$(4.7) \quad \psi_p(\mathbf{A}) \mathbf{V}_r = \mathbf{V}_r^+ \widehat{\mathbf{R}}_r,$$

where $\widehat{\mathbf{R}}_r$ is an upper triangular matrix of order $r \cdot b$.

Proof. From Lemma 4.1 it suffices to show that the QR factorization of

$$\psi_p(\mathbf{H}_m) [\mathbf{E}_1 \quad \cdots \quad \mathbf{E}_r] = \mathbf{W}_1 \widehat{\mathbf{R}}_r$$

for some upper triangular $\widehat{\mathbf{R}}_r$ of order $r \cdot b$. But this is a consequence of the link between the QR algorithm and nonstationary simultaneous iteration because

$$\psi_p(\mathbf{H}_m) = \mathbf{W} \mathbf{R}$$

is a QR factorization [37, p. 353]. The result follows from equating the initial $r \cdot b$ columns of this equality and letting $\widehat{\mathbf{R}}_r$ denote the leading submatrix of order $r \cdot b$ of $\mathbf{R}$. $\square$

The theorem allow us to link simultaneous iteration with an IRAM. If $p = 1$, $\psi_1(\lambda) = \lambda$, $r = 1$, and $b > 1$, then (4.7) becomes $\mathbf{A} \mathbf{V}_1 = \mathbf{V}_1^+ \widehat{\mathbf{R}}_1$ and simultaneous iteration is recovered. From (4.5) it follows that for this specific choice of polynomial an IRAM is simultaneous iteration in disguise because $\mathbf{V}_1^+$ is computed. Two remarks are in order.

First, we note that (4.3) through (4.6) define a restart of the Arnoldi reduction via a QR algorithm. An IRAM is a sequence of implicit restarts that are terminated when the partial Schur decomposition of interest is sufficiently well approximated. The restart is implicit because as Theorem 4.2 demonstrates, polynomials in $\mathbf{A}$ can be implicitly applied to $\mathbf{V}_m$ via an application to $\mathbf{H}_m$.

Second, if $p = m$, implicitly restarting with $m - 1$ shifts produces

$$(4.8) \quad \mathbf{A} \mathbf{U}_1^+ = \mathbf{U}_1^+ \mathbf{H}_1^+ + \mathbf{F}_1^+.$$

If $\mathbf{Q} \mathbf{R} = \mathbf{H}_1^+ - \tau_m \mathbf{I}$, then

$$(4.9) \quad (\mathbf{A} - \tau_m \mathbf{I}) \mathbf{U}_1^+ = \mathbf{U}_1^+ \mathbf{Q} \mathbf{R} + \mathbf{F}_1^+.$$

The right-hand side of (4.9) defines a new starting block of vectors (after orthogonalization) for a subsequent block Arnoldi reduction. Thus we can implicitly apply polynomials of degree m in $\mathbf{A}$. This bit of cleverness was first established in [3].

5. Convergence of an IRAM. Sorensen [34] gave some convergence results for an IRAM (blocksize of one). For nonsymmetric $\mathbf{A}$, a linear rate of convergence was given for a fixed $\psi_p(\cdot)$ per restart. He also showed that for symmetric $\mathbf{A}$, using the unwanted $m - k$ eigenvalues as shifts during each implicit restart resulted in convergence to k eigenvalues of $\mathbf{A}$.

Traditional convergence theory [13, 14, 27] for Arnoldi reductions investigates the quality of $\mathcal{K}_j^b(\mathbf{A}, \mathbf{U}_1)$ to approximate eigenvectors of $\mathbf{A}$ as j increases. However, with the connection between an IRAM and nonstationary simultaneous iteration in hand, a more elegant and powerful theory is possible. A comprehensive geometric convergence theory for the shifted QR algorithm is presented by Watkins and Elsner [40] within the more general framework of generic *GR algorithms*. A GR algorithm is a generalization of the QR algorithm where the QR factorization of $\mathbf{A} - \tau\mathbf{I}$ is replaced with a GR factorization, where $\mathbf{G}$ is a nonsingular matrix. The convergence theory is based on the idea that a GR algorithm is a nested sequence of nonstationary simultaneous iterations.

Theorem 5.1 in [40] discusses the convergence of nonstationary simultaneous iteration. Recall the notation established in sections 2 and 4. In addition, let $\phi_i(\cdot)$ denote $\psi_p(\cdot)$ (the polynomial in $\mathbf{A}$ implicitly applied) used at the i th restart of an IRAM.

THEOREM 5.1. *Let $\mathbf{A}$ be a simple matrix of order n . Let $r \cdot b \ll n$, where b and r are positive. Define the invariant subspaces $\mathcal{Z} = \text{Span}(\mathbf{x}_1, \dots, \mathbf{x}_{r \cdot b})$ and $\mathcal{U} = \text{Span}(\mathbf{x}_{r \cdot b + 1}, \dots, \mathbf{x}_n)$. Let $\Phi_i = \phi_i \cdots \phi_1$ and suppose that $\Phi_i(\lambda_j) \neq 0$ for $j = 1, \dots, r \cdot b$ and let*

$$(5.1) \quad \rho_i = \frac{\max_{j=r \cdot b + 1, \dots, n} |\Phi_i(\lambda_j)|}{\min_{j=1, \dots, r \cdot b} |\Phi_i(\lambda_j)|}.$$

If $\mathcal{V}$ is a subspace of dimension $r \cdot b$ satisfying $\mathcal{V} \cap \mathcal{U} = \{0\}$, then there exists a constant C such that

$$\text{dist}(\Phi_i(\mathbf{A})\mathcal{V}, \mathcal{Z}) \leq C\rho_i$$

for all i .

Proof. See Theorem 5.1 established in [40]. $\square$

We remark that the hypothesis on $\mathbf{A}$ is done only for notational and elaborative purposes. The paper [40] considers the more general case of an arbitrary matrix.

The distance between the subspaces [8, 10] $\mathcal{V}$ and $\mathcal{Z}$ may be shown to be equal to $\sqrt{1 - \sigma_{\min}^2(\mathbf{V}^H \mathbf{Z})}$ where the columns of $\mathbf{V}$ and $\mathbf{Z}$ provide an orthonormal basis for $\mathcal{V}$ and $\mathcal{Z}$ and $\sigma_{\min}(\cdot)$ denotes the minimum singular value. For increasing values of i , the approximating subspace $\Phi_i(\mathbf{A})\mathcal{V}$ aligns itself with $\mathcal{Z}$, and hence the distance between the two subspaces goes to zero.

An expression for the constant C is given by Watkins and Elsner. They show that

$$C = \|\mathbf{X}\| \|\mathbf{X}^{-1}\| \frac{\sqrt{1 - \sigma_{\min}^2(\mathbf{V}^H \mathbf{Z})}}{\sigma_{\min}(\mathbf{V}^H \mathbf{Z})},$$

where $\mathbf{X}$ is the matrix of eigenvectors for $\mathbf{A}$. The size of C is large whenever the eigenvectors form an ill-conditioned basis and/or the starting subspace $\mathcal{V}$ is nearly orthonormal to $\mathcal{Z}$.

Because of the relationship between an IRAM and nonstationary simultaneous iteration established in the last section, the $\text{dist}(\Phi_i(\mathbf{A})\mathcal{V}, \mathcal{Z}) \rightarrow 0$ and the eigenvalues

of the leading block of $\mathbf{H}_m$ of order $r \cdot b$ computed after i restarts are approximations to $\lambda_1, \dots, \lambda_{r \cdot b}$. The rate of convergence given by Theorem 5.1 is determined by how well the max-min problem (5.1) is solved. This problem cannot, in general, be solved. Section 6 discusses and reviews practical issues related to convergence.

6. Some practical issues. The question of determining a shift strategy that leads to a provable rapid rate of convergence is a difficult problem that continues to be researched. However, it is clear from Theorem 5.1 that a polynomial Φ_i that minimizes ρ_i for the discrete max-min problem (5.1) is required. Unfortunately, this polynomial is impossible to compute because the eigenvalues are not known. A common alternative is to replace the max-min problem posed on the discrete set of eigenvalues with a problem posed on a convex region containing the unwanted eigenvalues. Before we consider shifting schemes based on this approach, we review two simple shifting schemes.

A special case of an iteration for which the discrete max-min problem (5.1) can be solved is when $\phi_i(\lambda) = \lambda^i$. In this special case, $\rho_i = |\lambda_k/\lambda_{k+1}|^{i \cdot p}$ and by hypothesis on the ordering of the eigenvalues, $|\lambda_{k-1}/\lambda_k| < 1$, and so convergence to the dominant invariant subspace is assured.

More general nontrivial stationary iterations where $\Phi_i(\cdot) = \phi_1(\cdot)^i$ and ϕ_1 is a degree p polynomial converge at the linear rate

$$\frac{\max_{j=r \cdot b+1, \dots, n} |\Phi_i(\lambda_j)|}{\min_{j=1, \dots, r \cdot b} |\Phi_i(\lambda_j)|}$$

if $|\phi_1(\lambda_i)| \geq |\phi_1(\lambda_k)|$ for $i = 1, \dots, k$ and $|\phi_1(\lambda_k)| > |\phi_1(\lambda_i)|$ for $i = k+1, \dots, n$. This is a slight generalization of Theorem 5.1 in [34] easily established by Theorem 5.1 in the paper.

We now replace the max-min problem posed on the discrete set of eigenvalues with a problem posed on a convex region that contains the unwanted eigenvalues. A first step in reducing ρ_i is to bound the distance from the k dominant eigenvalues of $\mathbf{A}$ to the set of unwanted eigenvalues. If we assume that the dominant eigenvalues of $\mathbf{A}$ are all to the right (respectively, to the left) of λ_k , the denominator of (5.1) is bounded if all the τ_i 's can be enclosed in a convex region containing the unwanted eigenvalues not intersecting the region containing the dominant eigenvalues. If all the shifts lie in this latter region, then we can attempt to approximately solve

$$\rho_i = \frac{\max_{\mathcal{S}} |\Phi_i(\lambda)|}{\min_{\mathcal{W}} |\Phi_i(\lambda)|},$$

where $\mathcal{S}$ and $\mathcal{W}$ are, respectively, the unwanted and wanted regions.

When $\mathbf{A}$ is a real matrix, a standard approach is to set $\mathcal{S}$ to be an ellipse and then using the techniques described in [29, pp. 219–239] compute the roots of a Chebyshev polynomial. Saad shows that the per restart convergence to $\mathcal{V}$ is

$$\max_{i=1, \dots, k} \frac{\alpha + \sqrt{\alpha^2 - \gamma^2}}{\alpha_i + \sqrt{\alpha_i^2 - \gamma_i^2}},$$

where $\mathcal{S}$ is enclosed by an ellipse with center δ , focal distance γ , major axis α ; and γ_i , α_i are the focal distance and major axis of the ellipse including $\mathcal{S}$ and λ_i . The resulting rate of convergence can be far superior to the rate $|\lambda_{k-1}/\lambda_k|$ given by standard subspace iteration.

A second, more recent approach is the exact shift scheme proposed by Sorensen. This is the default scheme used by ARPACK [18]. An exact scheme uses a number of the unwanted eigenvalues of $\mathbf{H}_m$ for the $\psi_p(\cdot)$ implicitly applied per restart. The scheme is motivated by the observation that the scheme retains only the approximations to the wanted eigenvalues. As explained in [18, p. 71], the value of r is increased from the initial value of k as approximations to the eigenvalues of $\mathbf{A}$ satisfy the acceptance criterion. This has the effect of decreasing the numerator in (5.1) (see [35, 41] for similar conclusions). Unfortunately, there is no rigorous convergence theory giving a rate of convergence, but in practice the exact scheme has proven to be robust and gives slightly better results than using the zeros of Chebyshev polynomials.

Recent papers have elucidated the virtues of other restart polynomials. For non-Hermitian matrices $\mathbf{A}$, harmonic Ritz values [21, 20, 23, 32] and refined shifts [15] are interesting schemes that require further analysis. For symmetric eigenvalue problems, the roots of Leja polynomials [3, 5, 4] are successfully used for small m along with a deflation scheme that allows r to be increased from a value of one as satisfactory eigenvalue-eigenvector approximations emerge.

7. Numerical experiments. We present results for computing the 4 dominant eigenvalues and invariant subspace for the two-dimensional model convection-diffusion problem

$$-\Delta \mathbf{u}(x, y) + 4\mathbf{u}_x(x, y) + .5\mathbf{u}_y(x, y) = \lambda \mathbf{u}(x, y)$$

on the unit square $[0, 1] \times [0, 1]$ with zero boundary conditions. The problem is discretized by using centered finite differences where the mesh size $h = 1/(50 + 1)$ results in a matrix of order 2,500. Table 7.1 lists the results of some MATLAB experiments (IEEE double precision floating point arithmetic is used). “Subit” refers to simultaneous iteration along with a projection at each restart for unscrambling the Schur vector approximations. The purpose of these experiments is to demonstrate the dramatic difference in performance that can be achieved when iterating on subspaces of equal dimension.

Each computed dominant partial Schur decomposition of order 4 gave a residual no larger than 10^{-3} , and all experiments produced Ritz values that agreed to at least four significant digits. The first three lines of the table give results in terms of the number of restarts and matrix-vector products used when using $m \cdot b = 16$ vectors. The next five lines list the same output when $m \cdot b = 24$, and the final line indicates the total number of matrix-vector products needed when $b = 1$ and m is increased until the partial Schur decomposition is approximated. This final experiment gives an indication of the minimum number of matrix-vector products needed. When $b = 1$, IRAM uses the strategy for increasing r from k as the eigenvalues satisfy the acceptance criterion to a maximum of 8; for $b = 2$ and $b = 4$, the value of r stayed fixed at $m - 4$ and for $b = 3$ was fixed at $r = 5$.

The following are some other conclusions resulting from the experiments of Table 7.1:

- The simultaneous iteration results always performed an extra m matrix-vector products per restart needed for the projection step. This projection is not needed at every restart. Hence the minimum number of matrix-vector products needed is half the number listed.
- Decreasing the blocksize requires a larger value of acceptance tolerance to be used so that the four dominant eigenvalues are computed to four significant digits.

TABLE 7.1

Computing the partial Schur decomposition corresponding to the 4 dominant eigenvalues of a convection-diffusion matrix of order 2,500.

	k	m	Matvec	Restarts	b
Subit	4	1	17,664	552	16
IRAM	4	16	253	23	1
IRAM	4	8	288	35	2
Subit	4	1	16,800	350	24
IRAM	4	24	230	12	1
IRAM	4	12	264	16	2
IRAM	4	8	330	35	3
IRAM	4	6	376	45	4
IRAM	4	200	199	0	1

- If $\mathbf{A}$ is efficiently applied to the block of vectors, then the number of matrix-vector products should be divided by the blocksize. Hence simultaneous iteration may be feasible.
- Decreasing the acceptance tolerance and/or increasing k (number of eigenvalues to compute) hinders the efficiency of simultaneous iteration. This effect is less pronounced as b decreases. IRAM with a blocksize of one has an amazing ability to compute partial Schur decompositions with small backward error.

In summary, if storage considerations and/or the cost of orthogonalization prevent a large Arnoldi reduction from being computed, an IRAM does not require a substantial number more matrix-vector products. This was also noted by Morgan [22].

8. Conclusions. We showed that restarted Arnoldi methods are equivalent to nonstationary simultaneous iteration methods by demonstrating a relationship with implicitly restarted Arnoldi methods. Convergence theory and numerical experiments confirmed that $\Phi_i(A)\mathcal{K}_r^b(\mathbf{A}, \mathbf{U}_1)$ tends to the dominant invariant subspace of order $r \cdot b$ at a substantially faster rate than $\mathbf{A}^i\mathcal{K}_1^{r \cdot b}(\mathbf{A}, \mathbf{U}_1)$ when appropriate restarting polynomials are employed.

Acknowledgments. I thank Mark Embree, Steve Wright, and the referees for improvements to the initial manuscript. I also thank Chris Beattie for suggesting Lemma 4.1 to me during a workshop held in the spring of 1995.

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